

LIGHTHIEF

How we protect your solar when the grid asks you to stop exporting

A short guide for commercial & industrial clients in Cyprus

Lighthief Cyprus Ltd · HE 477423 · solarfarms.cy · May 2026

Your solar investment should keep working for your business — even when EAC (the electricity network operator) limits how much power you can send to the grid. We design and commission every system so you stay **legal, safe, and saving money** during those periods.

What is curtailment?

Sometimes the Cyprus electricity network is under pressure — usually on sunny days when many solar systems export at once. EAC can send a signal to your site to **stop feeding power into the grid** for a while.

On most commercial buildings, a small device (installed by EAC) receives that signal. Many older systems react by **turning solar off completely**. That means you lose the benefit of free daytime power for your factory, offices, cold rooms, or pumps.

Curtailment is normal and legal. The question is how your system responds — and that is where we help.

What we do for you

On Lighthief projects we use a **smart zero-export** setup (typically with a **Deye** inverter, the same brand many commercial sites trust worldwide). Instead of shutting down when EAC signals curtailment:



Solar keeps running

Panels stay on and produce power for **your building first**.



Nothing goes to the grid

Export is held at **zero** while curtailment is active — exactly what EAC requires.



You still save

Every kWh used on site is a kWh you did not buy from the grid at retail rates.



You stay compliant

No risky workarounds. Your connection agreement and inspections remain in order.

Without protection vs with Lighthief

Basic setup (what we avoid)

- EAC signal → inverter switches **off**
- No solar for your business during curtailment
- You buy more from the grid at full price
- Lost savings on every curtailed sunny hour

Lighthief zero-export protection

- EAC signal → export stops, **solar continues**
- Power goes to **your loads** only
- Lower electricity bills during those hours
- Designed, tested, and documented for EAC approval

How it works — in simple terms

1

We measure what leaves your site

Sensors on your main supply tell the inverter how much power is going to or from the grid.

2

When EAC asks for curtailment

The inverter automatically limits export to zero — it does not need someone on site to change settings.

3

Solar serves your business first

Offices, machinery, cooling, and other loads use the power. Only what you need and cannot use stays off the grid.

4

When the signal ends

Normal operation resumes — including export to the grid if you are on net billing and have surplus power.

What this means for your electricity bill

If you are on **net billing** with EAC Supply, your meter still records import and export each month.

- ✓ During curtailment you will not earn export credits for those hours — because you are not exporting.
- ✓ You **will** still cut your import from the grid by using your own solar on site.
- ✓ That on-site saving is the main financial benefit of our protection — especially for businesses with daytime consumption.

Important: Bypassing EAC's curtailment equipment or trying to export anyway can put your connection at risk. Our approach is the approved way to maximise your savings while respecting grid rules.

What we take care of — and what you do

Lighthief (design & delivery)

- Specify inverter and protection suited to your site
- Design zero-export control into the system from day one
- Coordinate with your licensed electrician for installation
- Test that curtailment behaves correctly before handover
- Provide documentation for EAC inspection where required

You (site owner)

- Do not modify or bypass EAC's curtailment device
- Keep normal maintenance and monitoring in place
- Contact us if alarms or export behaviour look unusual

Who this is for

This protection is especially valuable for **shops, warehouses, factories, hotels, and offices** with net billing or self-consumption — anywhere daytime use is high and curtailment would otherwise waste sunny hours.

Larger solar parks (typically above 120 kWp) use a different remote-control path; we apply the same principle — **limit export, protect production** — with equipment matched to the scale of the project.

Questions about curtailment on your site?

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